Sumit Mishra

PhD Candidate in Robotics & AI

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Research Profile

Deep learning researcher specializing in multimodal learning and vision-language foundation models with focus on real-world applications. Extensive experience in designing and evaluating vision-language models for zero-shot learning, cross-modal alignment, and explainable AI systems. Strong background combining multimodal deep learning research with deployment in safety-critical applications, demonstrating ability to bridge foundational research with product impact.

Research Interests

- Vision-Language / Multimodal Foundation Models
- Zero-shot, Few-shot, and weakly supervised Learning in Multimodal Contexts
- Cross-modal Knowledge Transfer and Alignment
- Explainable AI in Computer Vision and Multimodal Systems
- Applications in Safety-Critical Real-World Systems
- Perception Systems for Intelligent Agents

Education

- **Ph.D. in Robotics** 2023-2026
  Korea Advanced Institute of Science and Technology (KAIST)
  GPA: 3.85/4.3
  Thesis: "Responsible AI for Safer Roads: Advancing Driver Assistance Systems Using Multimodal Framework"
  Advisor: Prof. Dongsoo Har
  Focus: Multimodal learning, vision-language models, cross-modal fusion
- **M.S. in Robotics** 2021-2023
  Korea Advanced Institute of Science and Technology (KAIST)
  GPA: 4.0/4.3
  Thesis: "Multi-modal Data Usage for Smart City Mobility Services"
  Advisor: Prof. Dongsoo Har
  Focus: Multimodal sensor fusion, vision-language integration
- **B.Tech. in Electronics and Communication Engineering** 2012-2016
  Galgotias College of Engineering and Technology (AKTU)

Research Publications

Journal Papers

1. **Mishra, S.**, Kumar, H., Shyam, P., Har, D., and Ko, KW. "Zero-Shot Road Anomaly Detection Based on Text Corpus Knowledge and Segmentation-First Localization", *IEEE Transactions on ETCI* (Under review)
2. **Mishra, S.**, Mishra, M., Ko, KW., and Har, D. "Self-Supervised Visual Language Learning for Explainable Accident-Prone Feature Discovery", *IEEE Transactions on Intelligent Transportation Systems* (Under review)

3. Mishra, M., **Mishra, S.**, and Shin, H. "Cross-Modal Distillation for Real-time Wildfire Detection and Localization in Edge-Deployed Aerial Vehicles", *ISPRS Journal of Photogrammetry and Remote Sensing*, Elsevier, 2026 (IF: 12.9)

4. **Mishra, S.**, Kumar, H., Ko, KW., and Har, D. "IMUDistill: Knowledge Transfer for Enhancing Low Precision IMU Performance", *IEEE Signal Processing Letters*, 2026 (IF: 3.9)

5. **Mishra, S.**, Sengar, N., and Har, D. "A Secure, Blockchain-Enabled Vehicular Sensor Communication Protocol With Deep Learning-Assisted Anomaly Detection", *IEEE Intelligent Transportation Systems Magazine*, 2025 (IF: 5.5)

6. **Mishra, S.**, Mishra, M., Shyam, P., and Har, D. "TIME-VAD: Text-Informed Magnitude Enhancement Feature Learning for Vehicle Accident Detection", *IEEE Transactions on Intelligent Transportation Systems*, 2025 (IF: 9.1). Featured in KAIST Breakthrough.

7. Mishra, M., **Mishra, S.**, and Har, D. "Integrating Multi-sourced Sensor Data for Enhanced Traffic State Estimation", *IEEE Sensor Journal*, 2024 (IF: 4.3)

8. **Mishra, S.**, Rajendran, P K., Santos V, L F., and Har, D. "Sensing accident-prone features in urban scenes for proactive driving and accident prevention", *IEEE Transactions on Intelligent Transportation Systems*, 2023 (IF: 9.1). Featured in KAIST Breakthrough.

9. **Mishra, S.**, Singh, V., Gupta, A., Bhattacharya, D., and Mudgal, A. "Adaptive Traffic Signal Control in Developing Countries Using Parameters from Crowd-Source Data", *Transportation Letters*, Taylor & Francis, 2022 (IF: 3.598)

10. **Mishra, S.**, Singh, N., and Bhattacharya, D. "Application based COVID19 micro-mobility solution for safe and smart navigation in pandemics", *ISPRS International Journal of Geo-Information*, MDPI, 2021 (IF: 3.4)

11. **Mishra, S.**, Kushwaha, A., Aggrawal, D., Gupta, A. "Comparative emission study by real-time congestion monitoring for stable pollution policy on temporal and meso-spatial regions in Delhi", *Journal of Cleaner Production*, Elsevier, 2019 (IF: 11.072)

12. **Mishra, S.**, Bhattacharya, D., Gupta, A. "Congestion adaptive traffic light control and notification architecture using Google maps APIs", *Data*, 2018, 3, 67.

13. **Mishra, S.**, and Ahuja, B. "Optical parameters testing to redefine visibility for low cost transmissometer using channel modeling", *Optik-International Journal for Light and Electron Optics* 127.23, 2016 (IF: 3.1)

Conference Papers

1. **Mishra, S.**, Mishra, M., Kim, T., and Har, D. "RoadSafe: A Road Redesign Technique Achieving Enhanced Road Safety by Diffusion Model Inpainting", arXiv:2302.07440

2. Shyam, P., **Mishra, S.**, Yoon, K J., and Kim, K S. "Infra Sim-to-Real: An efficient baseline and dataset for Infrastructure based Online Object Detection and Tracking using Domain Adaptation", 33rd IEEE Intelligent Vehicles Symposium, 2022

3. Rajendran, P K., **Mishra, S.**, Santos V, L F., and Har, D. "RELMOBNET: A Robust Two-Stage End-To-End Training Approach for MOBILENETV3 Based Relative Camera Pose Estimation", European Conference on Computer Vision. Cham: Springer Nature Switzerland, 2022

4. **Mishra, S.**, Rajendran, P K., and Har, D. "Socially acceptable route planning and trajectory behavior analysis of personal mobility device for mobility management with improved sensing", 9th Robot Intelligence Technology and Applications, RiTA 2021

5. **Mishra, S.**, Bhattacharya, D., Chaturvedi, A., and Singh, N. "Assessing micro-mobility services in pandemics for studying 10-minutes cities concept in India using geospatial data analysis: an application", IWCTS, ACM SIGSPATIAL workshop 2022

6. **Mishra, S.** "Network Security Protocol For Constrained Resource Devices In Internet Of Things", 12th IEEE Indian International conference INDICON 2015

7. **Mishra, S.**, Singh, V., Bhattacharya, D. "Framework for freight truck return-trip assignment system", 11th IEEE IC3 2018 conference

8. **Mishra, S.**, Bhattacharya, D., Gupta, A., Singh, V R. “Adaptive traffic light cycle time controller using microcontrollers and crowdsource data of Google APIs for developing countries”, ISPRS Annals, SDSC-2018

9. Bhattacharya, D., Painho, M., **Mishra, S.**, and Gupta, A. “Mobile traffic alert and tourist route guidance system design using geospatial data”, ISPRS Archives, 2nd SDSC UDMS IEEE Conference, 2017

Patents

- Gupta, A. and **Mishra, S.** “System for spatio-temporal traffic data reporting for smart city services”, Indian Patent No.: 538566, Granted 17/05/2024.
  - Implemented in Hyderabad and Bangalore, India
  - Focuses on adaptive signaling technology
- Har, D., **Mishra, S.**, Mishra, M. “Real-time traffic monitoring and analysis system” Korean patent application no.: 10-2024-0103095 (Pending, KAIST support for Patent filing)
- Har, D., **Mishra, S.**, Mishra, M. “Accident probability prediction system”, Korean patent application no.: 10-2024-0087193 (Pending, KAIST support for Patent filing)

Work Experience

- **Remote Research Intern** 06/2025 - 11/2025
  Supervised by Dr. Pranjay Shyam, Applied Scientist II, Amazon Fulfillment Technologies & Robotics
  - Built zero-shot road anomaly detection achieving 92.42% accuracy using CLIP, outperforming VLMs (LLaVA, BLIP) by 16-22% with 15-25× smaller model size
  - Designed Triple-CLS architecture for GPT2-based safety notification generation without paired training data
  - Conducted systematic foundation model benchmarking (CLIP variants, BLIP, LLaVA) analyzing accuracy-latency-size trade-offs for edge deployment
  - Contributed TIME-VAD framework (IEEE T-ITS): 94.44% accuracy on DoTA dataset, 9.74% over previous SOTA
- **Graduate Research Assistant** 03/2021 - Present
  Vehicular Intelligence Lab, KAIST
  - Developed VisText-Net (IEEE T-ITS submission): CLIP-based concept bottleneck model achieving 11.1% improvement in explainability for accident hotspot classification
  - Created TIME-VAD framework (IEEE T-ITS): Text-informed accident detection achieving 94.44% accuracy on DoTA dataset, outperforming SOTA by 9.74%; featured in KAIST Breakthrough
  - Built zero-shot anomaly detection system with 92.42% accuracy using CLIP, outperforming VLMs by 16-22% with 15-25× smaller model size
  - Co-developed lightweight wildfire detection for edge-AI (ISPRS JPRS): 90.96% accuracy with only 2.99 GFLOPS via thermal-to-optical knowledge distillation
  - Deployed accident-prone feature notification system with Deloitte India (Lucknow region); featured in KAIST Breakthrough
  - Achieved state-of-art for accident detection task on multiple datasets: 95.38% (CCD), 84.43% (DAD), 94.44% (DoTA); demonstrated cross-domain generalization with 95.50% accuracy
  - Led team consisting of personnel from industry and academia for deployed safe navigation portal for COVID-19 era safe routing with Deloitte, GIZ, and University of Edinburgh, securing £11,500 grant and demonstrating cross-organizational research impact
  - Published 8+ papers in IEEE journals (IF: 4.3-8.5) and Conferences on multimodal sensing, secure vehicular communication, and sensor fusion
- **Research Consultant** 06/2018 - 02/2021
  Learnogether Technologies PVT. LTD.
  - Led end-to-end AI system design for smart city clients (professionals and academicians), managing projects from conception through deployment with integration into existing infrastructure

  - Developed drowsy driver detection and eye tracking systems using computer vision, deployed for commercial transportation safety applications
  - Collaborated with **Deloitte India** on adaptive traffic signaling implementation in Gurugram, contributing to system design and technical architecture for large-scale urban deployment
  - Provided technical consultancy for toll plaza congestion tracking and submitted proposals for congestion monitoring and visibility control systems

- **Research Scholar** 08/2016 - 06/2018
  Galgotias College of Engineering and Technology (Galgotias-Director's letter)
  - Collaborated with Dr. Ankit Gupta (IIT BHU) and Dr. Devanjan Bhattacharya (Univ. of Edinburgh) on joint research projects resulting in patent grant (Indian Patent No.: 538566) for spatio-temporal traffic data system
  - Developed real-time congestion mapping methodology for Delhi region similar to Google Environmental Insights, leading to publication in Journal of Cleaner Production (IF: 11.072)
  - Conducted hardware/software POC development including IoT network security protocols, traffic light control systems, and low-cost visibility measurement technology for smart city applications
  - Published 4 papers on adaptive traffic systems and geospatial data analysis, establishing foundation for data-driven urban mobility research

Technical Skills

- **Vision-Language Foundation Models:** CLIP, DINO, Florence, BLIP, LLaVA | Concept bottleneck models | Zero-shot/few-shot learning | Prompt engineering | Cross-modal alignment
- **Multimodal Learning:** Cross-modal knowledge distillation (thermal→optical, vision→text) | Contrastive learning (SimCLR, MoCo, text-anchor-based) | Weak Supervised learning | Multi-instance learning under weak supervision
- **Deep Learning & Computer Vision:** PyTorch (Primary), TensorFlow | Transformers | Vision Transformers (ViT, DeiT) | Attention mechanisms | Dilated temporal convolutions | OpenCV, Detectron2, MMDetection | CAM/Grad-CAM for explainability | Self-supervised learning | Domain adaptation | Model compression | Edge-AI optimization
- **Programming & Tools:** Python (Expert - 8+ years) | CLIP, Hugging Face | Git, Docker, Weights & Biases, TensorBoard, Jupyter

Academic Service

Reviewer

- **Conferences:** IEEE Intelligent Vehicles Symposium (IV), Conference on Human Factors in Computing Systems (CHI)
- **Journals:** IEEE Transactions on Intelligent Transportation Systems, IEEE Sensors Journal, IEEE Access, Array, Mobile Networks and Applications, Multimedia Tools and Applications, Scientific Reports, Qeios, Cluster Computing, Peer-to-Peer Networking and Applications

Talks

- Multi-Modal Data Usage for Smart City Use Cases Especially Traffic Surveillance, GCET
- AI/ML Applications in Road Safety and Traffic, IIT BHU

Awards and Grants

- Full-funding KAIST scholarship for MS and PhD in Robotics 2021-Present
- Project fund grant of 12.5 lakh INR from MSME, India 2018
- Fund grant of £11500 from University of Edinburgh for "MobilitySafe" project 2020

Academic Coursework

- **Core AI & Deep Learning:** Advances in Convolutional Neural Networks, Deep Learning, Pattern Recognition, Introduction to Artificial Intelligence, Artificial Intelligence and Machine Learning
- **Computer Vision:** Computer Vision, Advanced Visual Recognition
- **Robotics & Perception:** Mobile Robotics, Wireless Link Analysis
- **Domain Applications:** Transportation Infrastructure System

Project Media Coverage

- **Accident prone-feature detection for ADAS**
  - Featured in KAIST Breakthrough
- **Intelligent Traffic Data Fusion System**
  - Covered by The Times of India (National English Newspaper)
  - Featured in Amar Ujala (National Hindi Newspaper)

References

- **Prof. Dongsoo Har**
  Professor, The CCS Graduate School of Mobility
  Korea Advanced Institute of Science and Technology
  Email: dshar@kaist.ac.kr
  Website
- **Dr. Pranjay Shyam**
  Applied Scientist II
  Amazon Fulfillment Technologies & Robotics, Seattle, USA
  Email: pranjayy@amazon.com
  LinkedIn
- **Dr. Luiz Felipe Vecchietti**
  Senior Researcher
  Max Planck Institute for Security and Privacy (MPI-SP), Germany
  Email: felipe.vecchietti@mpi-sp.org
  Website
- **Prof. Ankit Gupta**
  Professor and MoRTH Chair Professor, Transport Engineering Section
  Department of Civil Engineering, IIT (BHU)
  Email: ankit.civ@iitbhu.ac.in
  Website
- **Dr. Devanjan Bhattacharya**
  Marie Skłodowska-Curie Actions (MSCA) TRAIN@ED Fellow
  School of Law and School of Informatics, University of Edinburgh
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  Website

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